

Technical & Theoretical Documentation

Customer Credit Risk Assessment & End-to-End Data Preprocessing Pipeline

PART A: CONCEPTUAL FOUNDATION

1. Pipeline Architecture & Risk Modeling Philosophy

Credit risk assessment relies on preprocessing structured customer metrics (income, loan amounts, credit scores) to ensure high predictive fidelity. The data preparation workflow standardizes input feature distributions, mitigates bias from missing observations, and transforms raw metrics into predictive leverage points for financial risk modeling.

PART B: DATA ACQUISITION & MULTI-SOURCE INGESTION

Modern credit risk assessment aggregates transactional, behavioral, and macroeconomic inputs from heterogeneous data stores:

Data Source	Target Attributes	Technical Purpose
CSV Engine	customer_id, loan_amount, annual_income	Primary batch transaction records and baseline customer profiles.
JSON Metadata	education_level, employment_type	Semi-structured demographic features and ordinal risk indicators.
SQL Repositories	repayment_history, default_status	Relational historical records providing target classification ground truth.
REST API Ingestion	inflation_rate, prime_interest_rate	External economic indicators providing contextual credit market benchmarks.

PART C: DATA UNDERSTANDING & MISSING VALUE IMPUTATION

1. Missing Data Mechanisms

MCAR (Missing Completely at Random): Missingness is entirely independent of observed and unobserved data.

MAR (Missing at Random): Missingness depends systematically on observed attributes (e.g., lower income reporting).

MNAR (Missing Not at Random): Missingness directly depends on the unobserved missing value itself.

2. Imputation Methodologies

Simple Imputer: Replaces missing numerical values with column Mean or Median, and categorical values with Mode.

KNN Imputer: Estimates missing values by computing Euclidean distance across k -nearest neighbors in feature space.

MICE (Iterative Imputer): Models each feature with missing values as a function of other features in a chained regression loop.

PART D: OUTLIER DETECTION & TREATMENT

Extreme financial values (e.g., ultra-high loans or unusual incomes) can severely distort machine learning algorithm weights. Robust outlier detection isolates real structural signals from noisy anomalies.

1. Interquartile Range (IQR) Method

Uses non-parametric quantiles ($Q1$ at 25th percentile, $Q3$ at 75th percentile) to identify extreme boundary deviations.

$$IQR = Q3 - Q1$$
$$Lower\ Bound = Q1 - (1.5 \times IQR), \quad Upper\ Bound = Q3 + (1.5 \times IQR)$$

2. Z-Score Normalization Outlier Bound

Measures how many standard deviations (σ) an observation (X) lies away from the population mean (μ).

$$Z = (X - \mu) / \sigma$$

3. Winsorization & Capping

Instead of dropping critical financial records, extreme values outside designated upper and lower percentile thresholds (e.g., 1st and 99th percentiles) are capped to boundary values, preserving sample size while eliminating variance distortion.

PART E: FEATURE ENGINEERING & CATEGORICAL ENCODING

Machine learning models require mathematical representations of categorical and temporal information:

- **Ordinal Encoding:** Maps ordered categorical levels to integer ranks (e.g., Secondary = 0, Graduate = 1, Post-Graduate = 2) while retaining structural rank hierarchy. Configured with `handle_unknown='use_encoded_value'` for production stability.
- **One-Hot Encoding:** Converts nominal non-ordered attributes into sparse binary vectors (0/1 indicators), avoiding artificial rank bias.
- **Discretization & Binning:** Groups continuous metrics into discrete buckets via Equal Width, Quantile, or K-Means clustering to capture non-linear relationships.

PART F: FEATURE SCALING & NON-LINEAR TRANSFORMATIONS

Scaling standardizes metric ranges so feature magnitude does not dominate gradient updates during model optimization.

Scaling / Transformation	Mathematical Formulation	Primary Use Case
Standardization (Z-Score)	$X_{scaled} = (X - \mu) / \sigma$	Centers data at mean 0 with variance 1. Ideal for algorithms assuming Gaussian inputs.
Min-Max Scaling	$X_{scaled} = (X - X_{min}) / (X_{max} - X_{min})$	Bounds features within range [0, 1]. Preserves exact zero entries in sparse matrices.
Robust Scaling	$X_{scaled} = (X - Q2) / IQR$	Uses Median (\$Q2\$) and IQR. Highly resilient against remaining outlier values.
Log Transformation	$Y = \log(1 + X)$	Reduces severe right-skewness in positive financial variables like annual income.
Power Transformer	Box-Cox & Yeo-Johnson	Stabilizes variance and normalizes non-Gaussian error terms across variables.

PART G: DOMAIN-SPECIFIC FEATURE CONSTRUCTION

1. Debt-to-Income (DTI) Ratio

Measures credit commitment relative to earning capacity. A primary driver in credit default prediction.

$$DTI\ Ratio = Total\ Loan\ Amount / Annual\ Income$$

2. Spending-to-Income Ratio

Evaluates net discretionary capacity by assessing total monthly expenditures against gross earnings.

$$Spending-to-Income = Monthly\ Expenditures / (Annual\ Income / 12)$$

PART H: FINAL PIPELINE DELIVERABLE & ML READINESS

The complete pipeline produces `final_customer_credit_risk_cleaned.csv`. By combining automated MICE imputation, capped outliers, ordinal/one-hot encoding, and standard scaling, the output dataset is fully numeric, zero-missing, mathematically scaled, and ready for immediate deployment in machine learning model training.